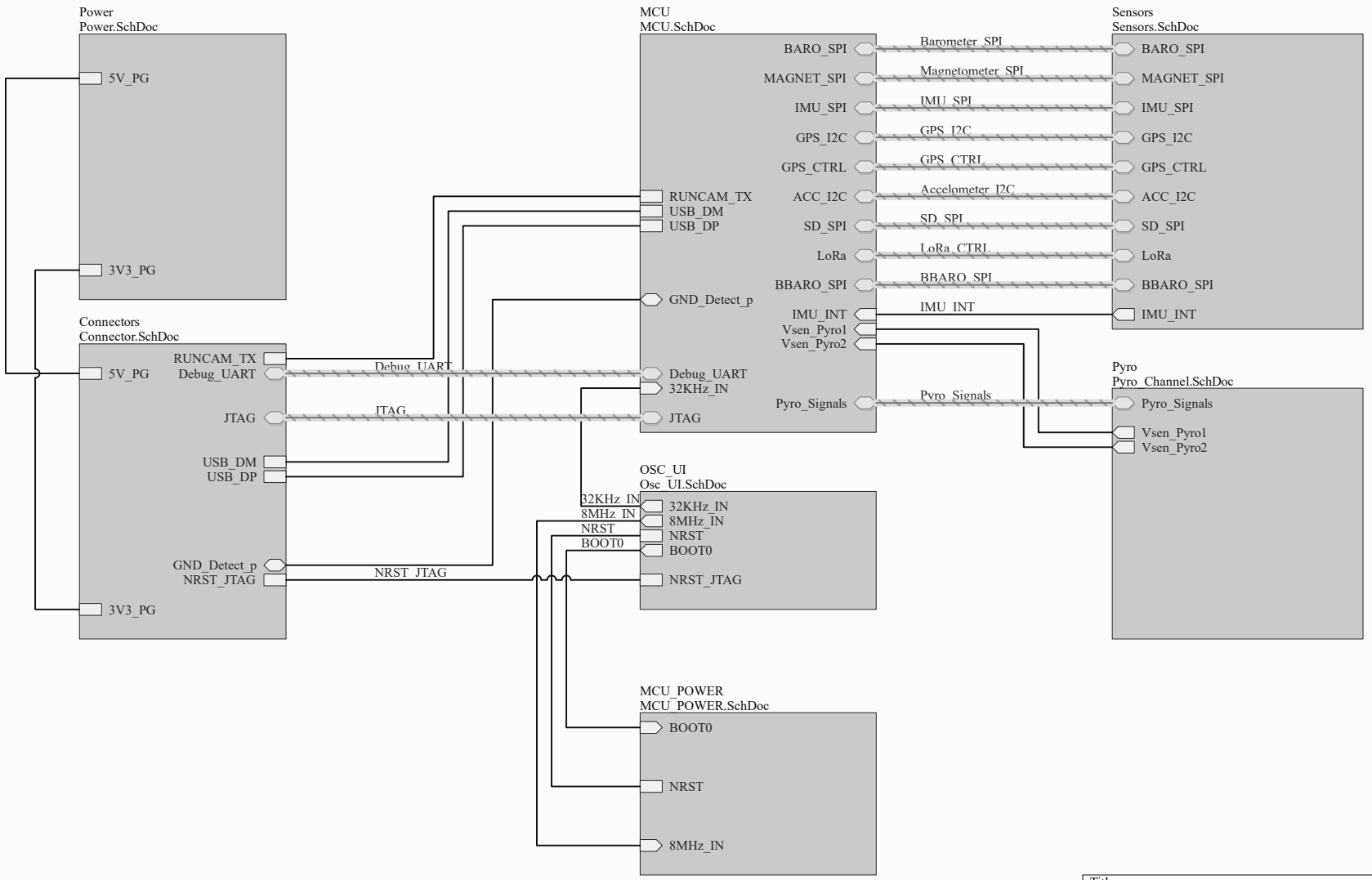
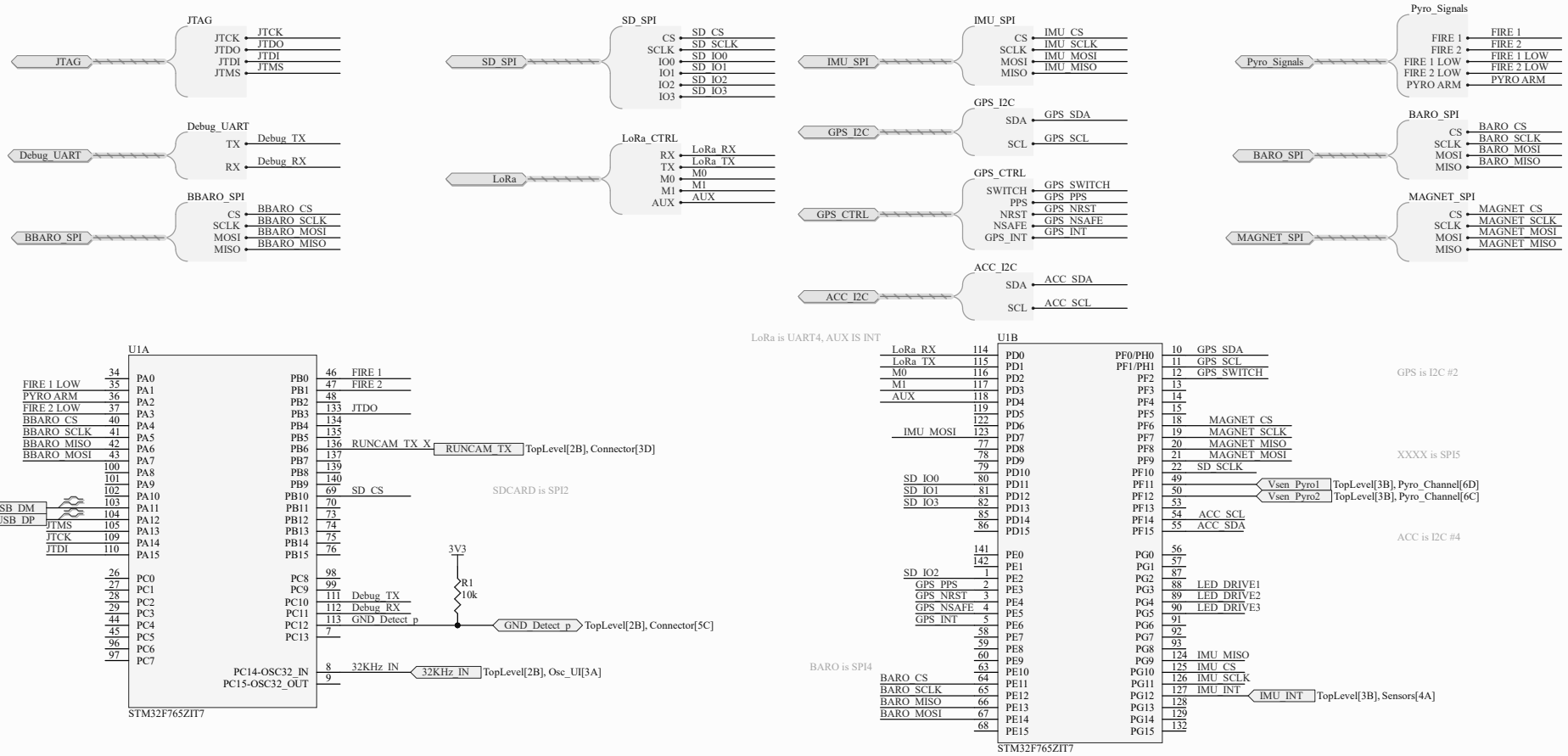
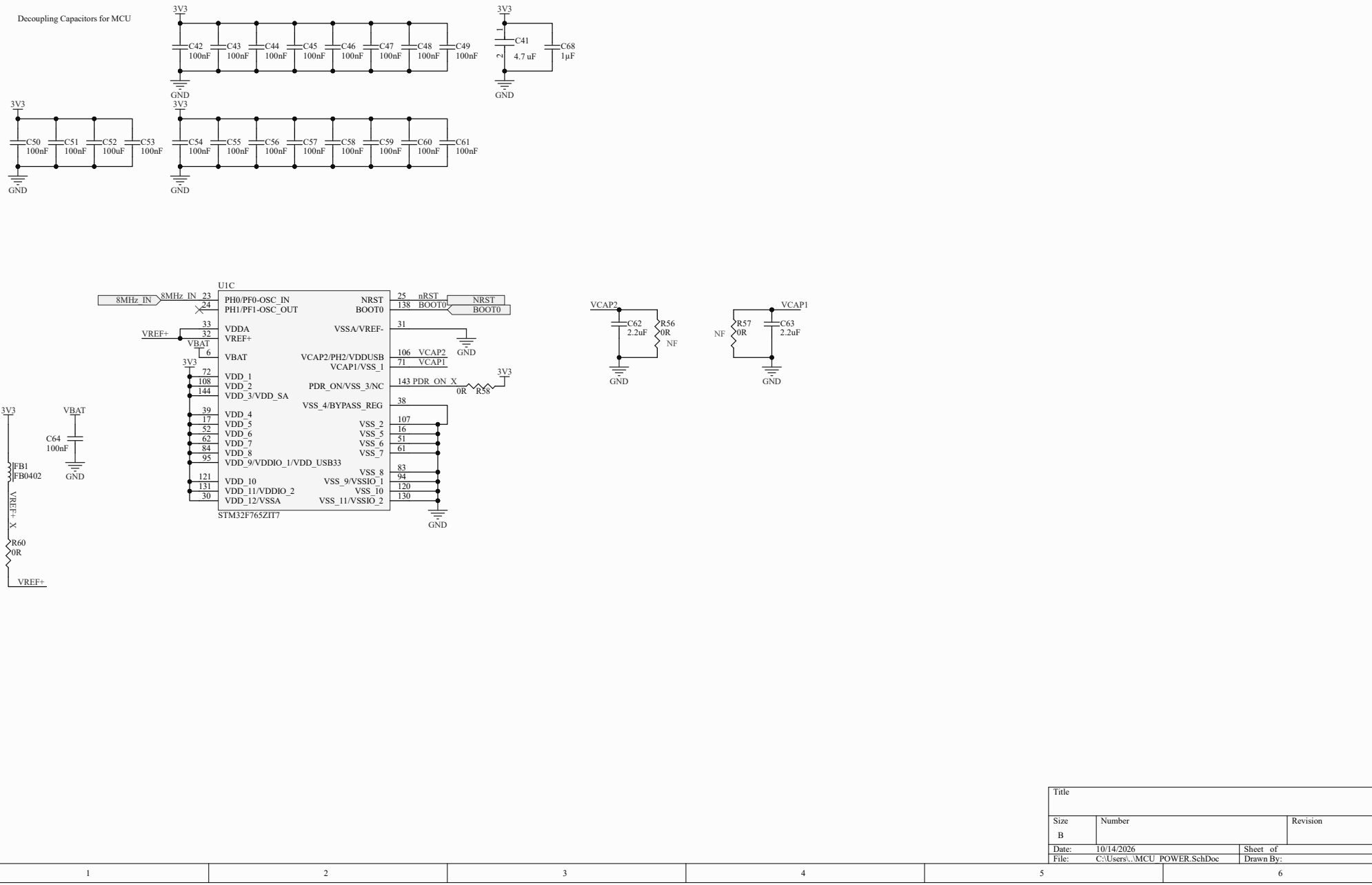


Title		
Size: A4	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\TopLevel.SchDoc		Drawn By:



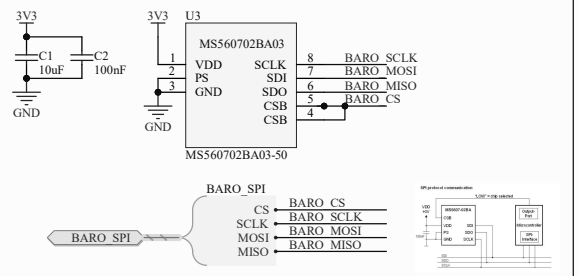
Title		
Size: B	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\MCU.SchDoc		Drawn By:



Title			
Size: B	Number:		Revision:
Date: 10/14/2026		Sheet of	
File: C:\Users\..\MCU_POWER.SchDoc		Drawn By:	

Barometer

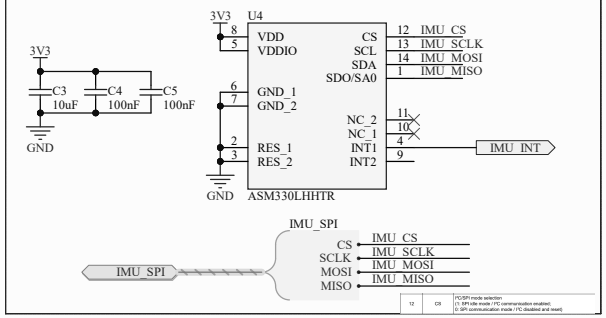
20MHz max clock



IMU

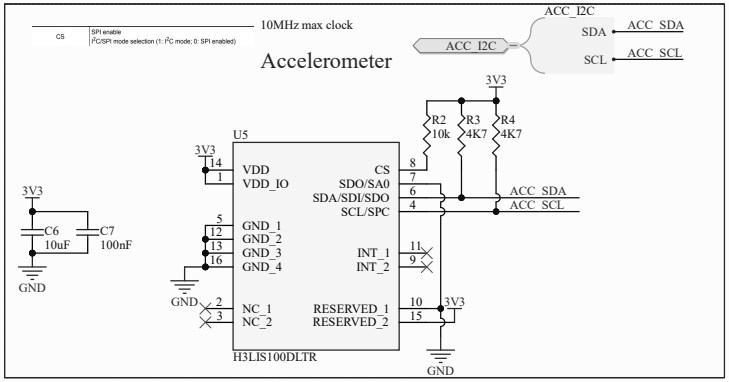
10MHz max clock

double check decoupling, INT1/2 Floating?



Accelerometer

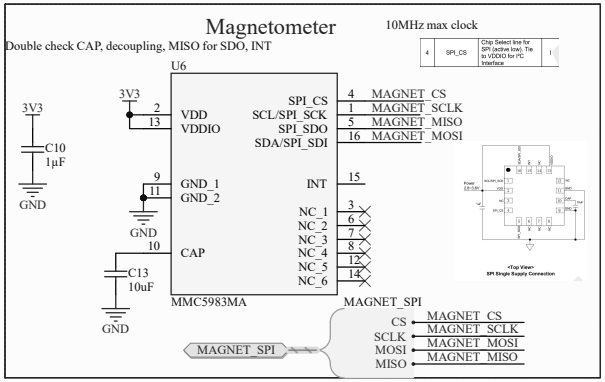
10MHz max clock



Magnetometer

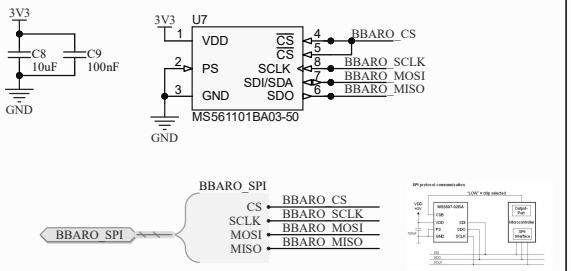
10MHz max clock

Double check CAP, decoupling, MISO for SDO, INT



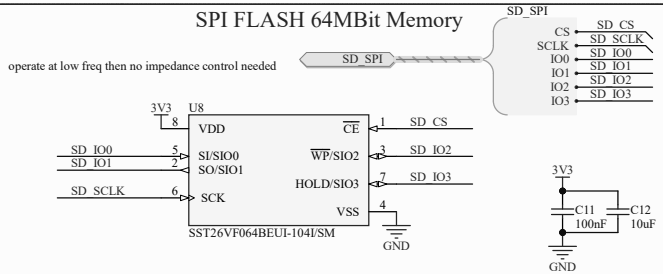
Backup Barometer

20MHz max clock



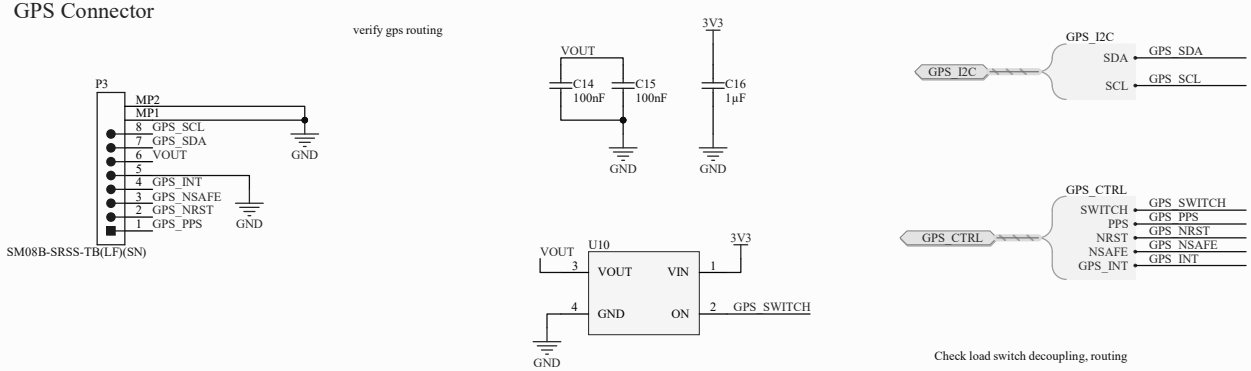
SPI FLASH 64MBit Memory

operate at low freq then no impedance control needed



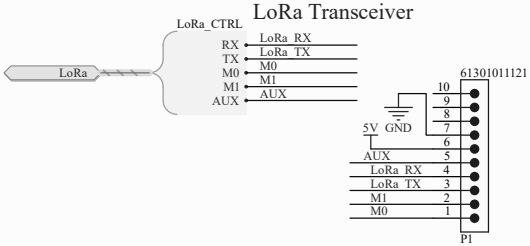
GPS Connector

verify gps routing



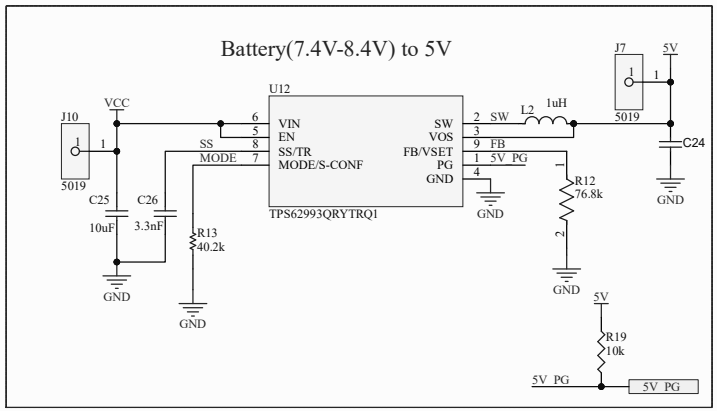
Check load switch decoupling, routing

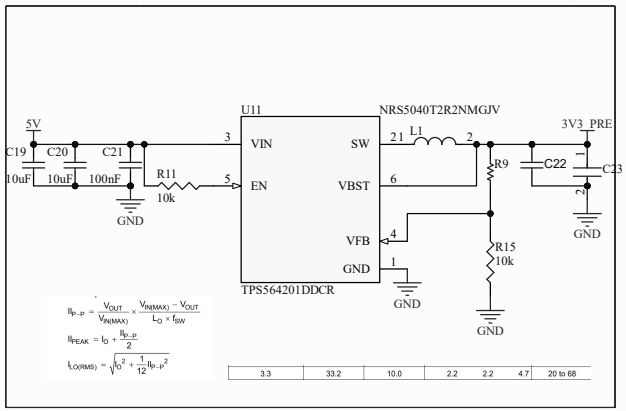
LoRa Transceiver

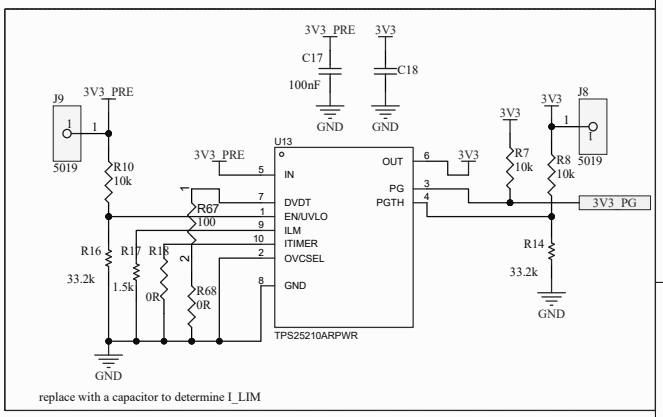


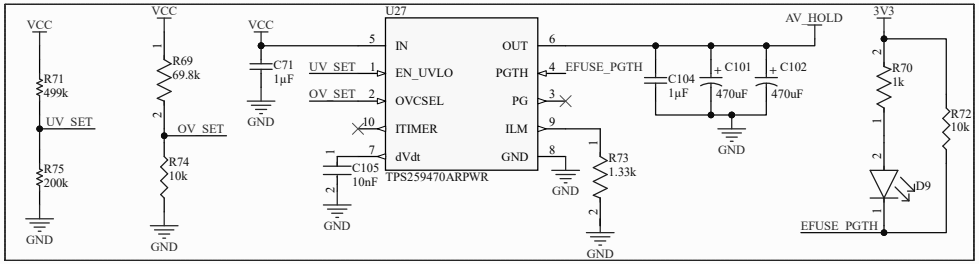
verify pins, specifically rx going to tx on chip, vice versa

Title		
Size: B	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\Sensors.SchDoc		Drawn By:









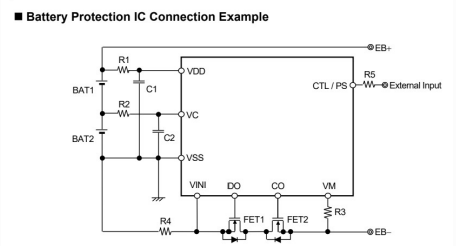
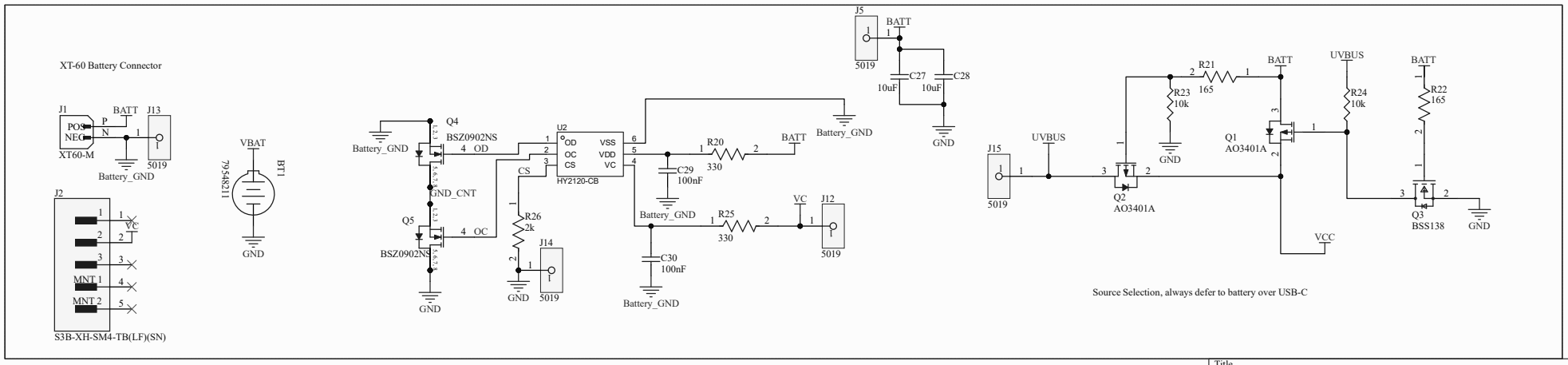


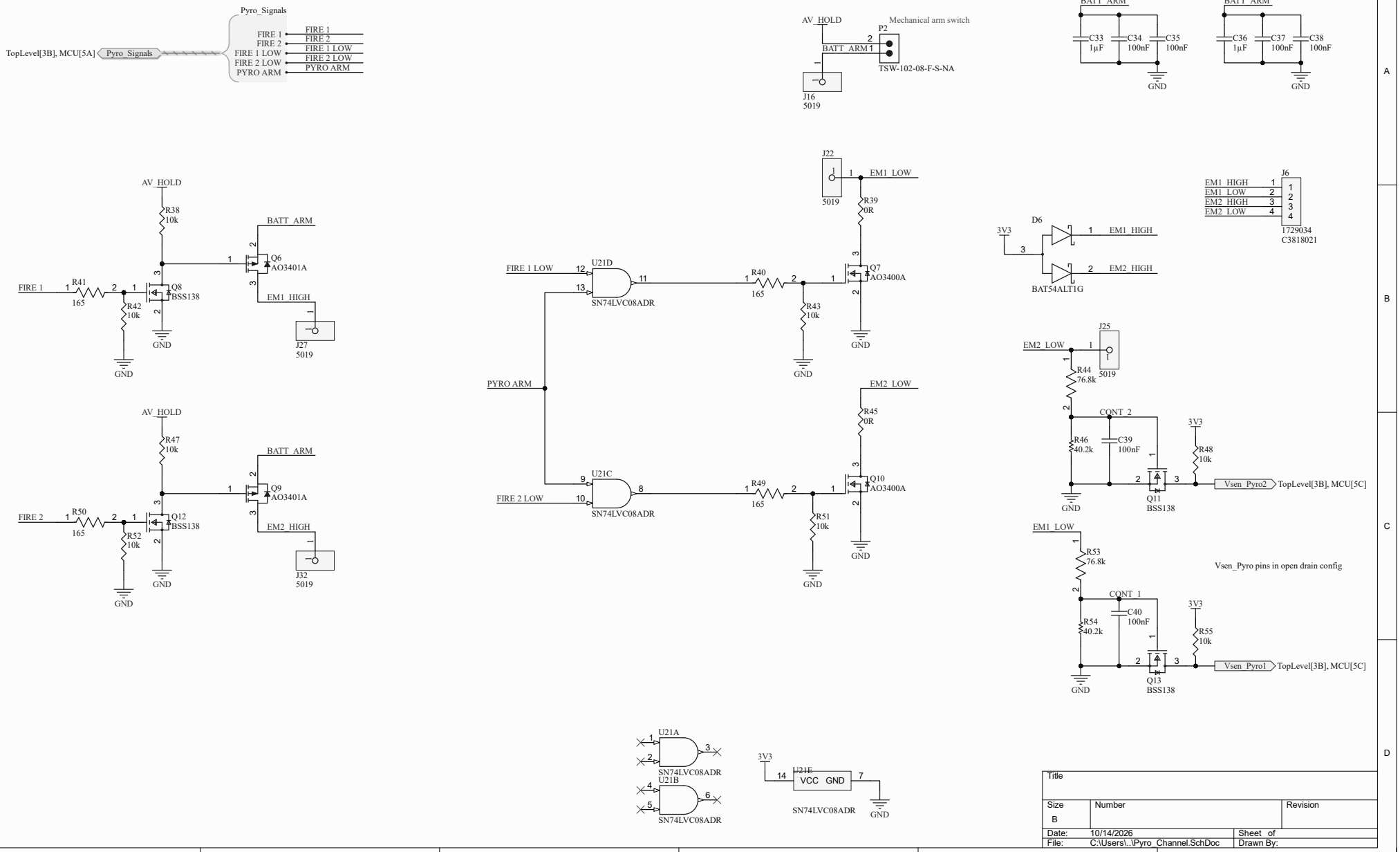
Table 19 Constants for External Components

Symbol	Part	Purpose	Min.	Typ.	Max.	Remark
FET1	N-channel MOS FET	Discharge control	–	–	–	Threshold voltage ≤ Overdischarge detection voltage*1.
FET2	N-channel MOS FET	Charge control	–	–	–	Threshold voltage ≤ Overdischarge detection voltage*1
R1, R2	Resistor	ESD protection, For power fluctuation	100 Ω	100 Ω	150 Ω*2	–
C1, C2	Capacitor	For power fluctuation	0.068 µF	0.1 µF	1.0 µF	–
R3	Resistor	ESD protection, Protection for reverse connection of a charger.	300 Ω	1.0 kΩ	1.5 kΩ	–
R4	Resistor	Overcurrent detection	–	1 mΩ	–	–
R5	Resistor	CTL / PS pin input protection	–	1 kΩ	–	–

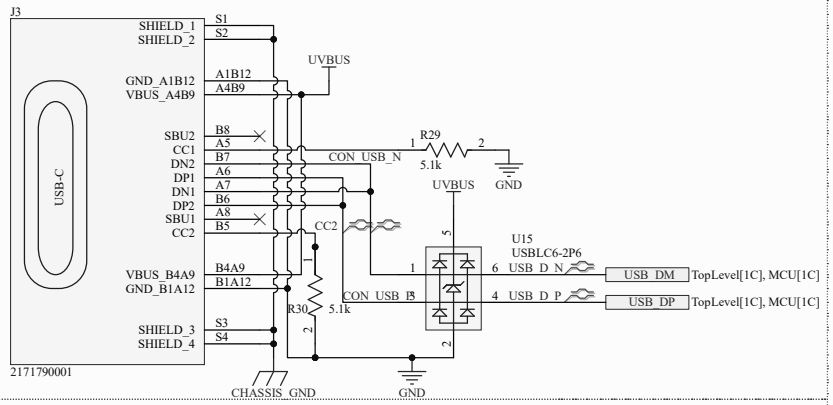
*1. If a FET with a threshold voltage equal to or higher than the overdischarge detection voltage is used, discharging may be stopped before overdischarge is detected.
*2. Accuracy of overcharge detection voltage is guaranteed by R1 = 100 Ω. Connecting resistors with other values will worsen the accuracy.



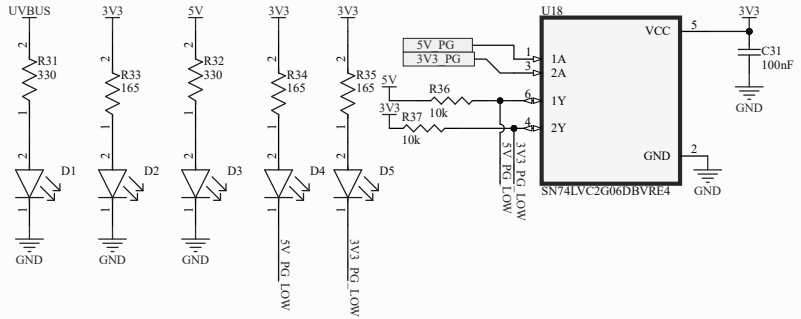
Title		
Size: B	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\Power.SchDoc		Drawn By:

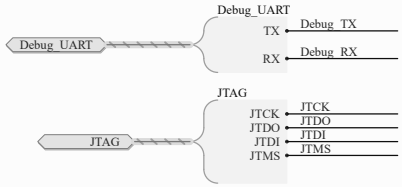

Pyro_Signals
FIRE 1
FIRE 2
FIRE 1 LOW
FIRE 2 LOW
PYRO ARM
TopLevel[3B], MCU[5A]
Pyro_Signals
AV_HOLD
Mechanical arm switch
P2
BATT_ARM
TSW-102-08-F-S-NA
J16
5019
BATT_ARM
C33 1µF
C34 100nF
C35 100nF
C36 1µF
C37 100nF
C38 100nF
GND
J22
EM1_LOW
R39 0R
J6
EM1_HIGH
EM2_HIGH
EM2_LOW
1729034
C3818021
D6
3V3
BAT54ALT1G
FIRE 1 LOW
U21D
SN74LVC08ADR
R40 165
R43 10k
Q7 AO3400A
PYRO ARM
J25
EM2_LOW
R44 76.8k
CONT_2
R46 40.2k
C39 100nF
R48 10k
Q11 BSS138
Vsen_Pyro2
TopLevel[3B], MCU[5C]
AV_HOLD
R38 10k
BATT_ARM
Q6 AO3401A
FIRE 1
R41 165
Q8 BSS138
R42 10k
EM1_HIGH
J27
5019
R45 0R
FIRE 2 LOW
U21C
R49 165
R51 10k
Q10 AO3400A
R47 10k
Q9 AO3401A
FIRE 2
R50 165
Q12 BSS138
R52 10k
EM2_HIGH
J32
5019
EM1_LOW
R53 76.8k
CONT_1
R54 40.2k
C40 100nF
R55 10k
Q13 BSS138
Vsen_Pyro1
Vsen_Pyro pins in open drain config
U21A
U21B
U21E
VCC GND
Title
Size B
Number
Revision
Date: 10/14/2026
Sheet of
File: C:\Users\...\Pyro_Channel.SchDoc
Drawn By:

USB PORT

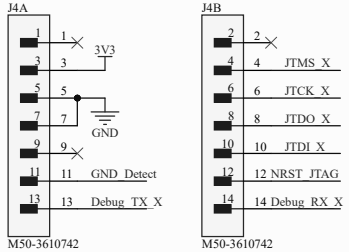


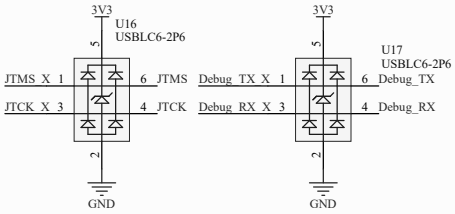
DEBUG LEDS

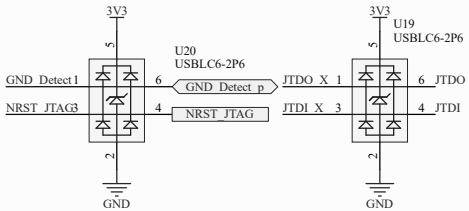




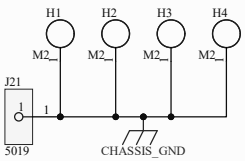
Header according to the STDC14 Connector used by MB1441

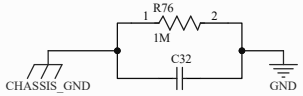




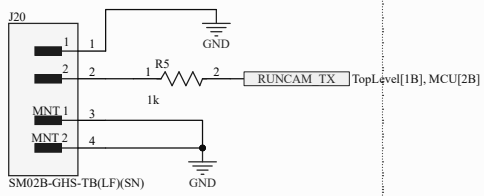


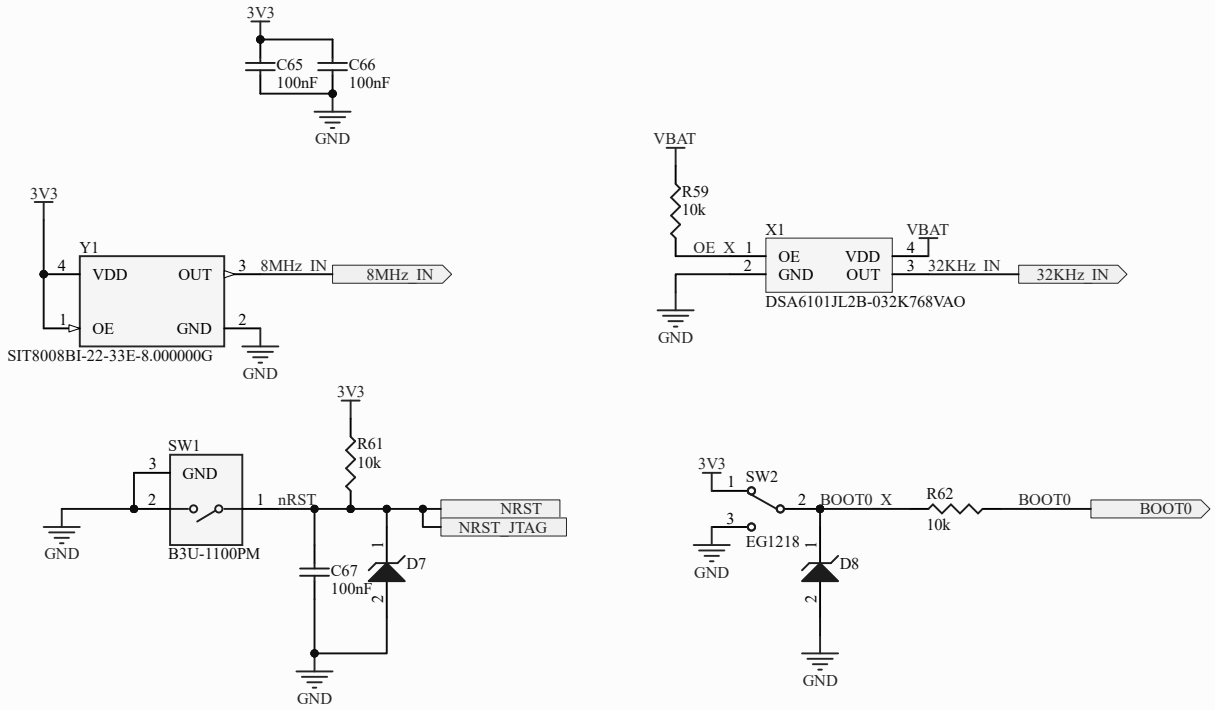
MOUNTING HOLES





RUN CAM CONNECTOR





Title		
Size: A4	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\Osc_UI.SchDoc		Drawn By:

4-Layer Impedance Control Stackup

Thickness

0.8mm | 1.0mm | 1.2mm | **1.6mm** | 2.0mm

Outer Copper Weight

1oz | 2oz

inner Copper Weight

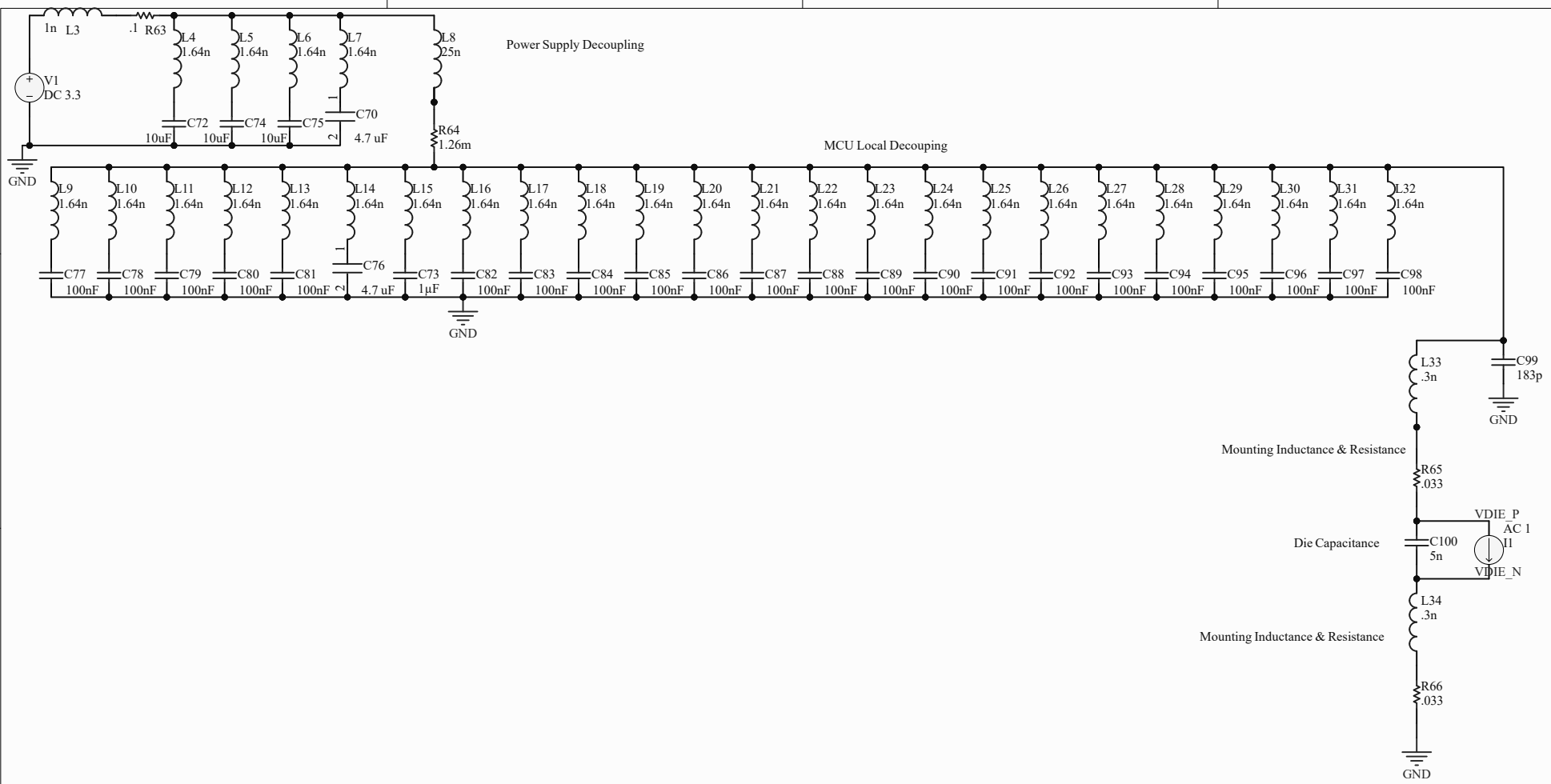
0.5oz | 1oz | 2oz

1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	7628 Dielectric Constant of 4.4
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	7628 Dielectric Constant of 4.4
Bottom Layer	Copper	0.035mm	



Title		
Size: B	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\Stackup.SchDoc		Drawn By:



Title		
Size A4	Number	Revision
Date: 10/14/2026		Sheet of
File: C:\Users\..\PDN_3.3V_Rail.SchDoc		Drawn By: